

***Do successful sports programs raise
graduation rates?***

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1- Introduction

For this exam, I performed a MLR analysis on the NCAA graduation rates dataset that was downloaded. The data comes from the 1996-1999 editions of US News' "Best Colleges in America" and the US Department of Education. It was originally presented by Mangold, Bean, and Adams (2003) in the journal article *The Impact of Intercollegiate Athletics on Graduation Rates Among Major NCAA Division I, II, Universities*.

I focused on evaluating the graduation rates among Major NCAA Division I universities, considering 19 variables that could potentially influence these rates.

Table 1 details 19 x-variables and their names in the dataset.

Variable #	Title in Dataset	Description
X_1	top10	% students in top 10% of high school
X_2	act25	ACT composite 25th percentile
X_3	oncampus	% living on campus
X_4	ft_grad	% 1st time undergrads
X_5	size	Total enrollment/100
X_6	tateach	% courses taught by TAs
X_7	bbindex	Composite basketball ranking
X_8	tuition	In-state tuition/1000
X_9	board	Room & board/1000
X_{10}	attend	Avg basketball home attendance
X_{11}	full_sal	Full professor salary
X_{12}	sf_ratio	student/faculty ratio
X_{13}	white	% white students
X_{14}	ast_sal	Assistant professor salary
X_{15}	pop	Population of city
X_{16}	phd	% faculty with PhD

X_{17}	accept	Acceptance rate
X_{18}	l_pct	% with loans
X_{19}	outstate	% out of state
Y	grad_rate	Avg 6 year grad rate for 1996, 1997, 1998

Table 1: Variable Names, Titles in Dataset, Descriptions

2 – Evaluating the Normal Distribution of the Y-Data

To ensure the validity of our linear model, it is crucial that the response variable, **grad_rate**, follows a normal distribution. To assess this, I created a histogram and a Q-Q plot for **grad_rate**, which are standard methods for evaluating normality.

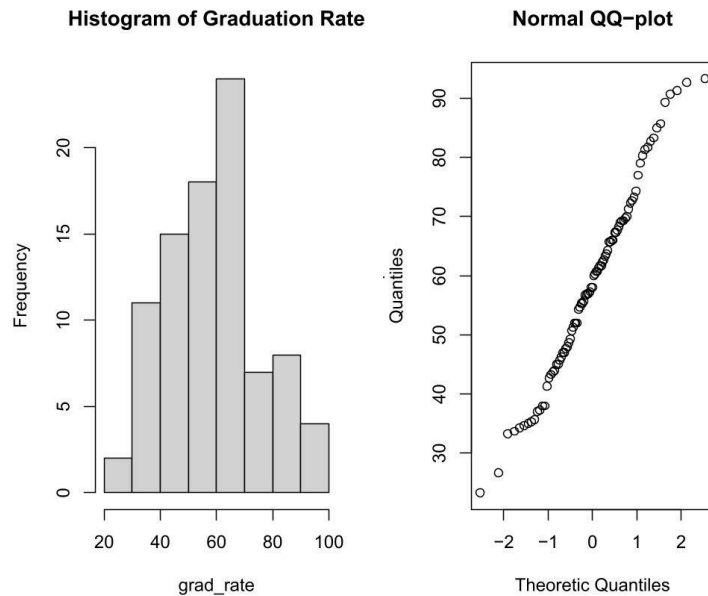


Figure 1: Normality checking of the Y-Data Using Histogram and Normal QQ-plot

Based on the histogram, the data is skewed right, suggesting the data is not normal. Additionally, the normal QQ-plot is not a straight line and the ends of the line are severely curved. If the data were normal, the normal QQ-plot would show a straight line. These two graphs are evidence that our data is not normally distributed. Therefore, we need to perform a Box-Cox transformation.

3 – Box-Cox Transformation

The Box-Cox method is a statistical technique used for transforming non-normally distributed data into approximately normal distributions. This transformation involves raising the data to a power λ which is determined through optimization to achieve normality or stabilize variance.

$$y_i^{(\lambda)} = \begin{cases} \frac{y_i^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \ln(y_i) & \text{if } \lambda = 0, \end{cases}$$

To transform the response variable **grad_rate**, I utilized the Box-Cox function in R to generate a Box-Cox plot. This plot helps identify the optimal λ value that maximizes the log-likelihood function.

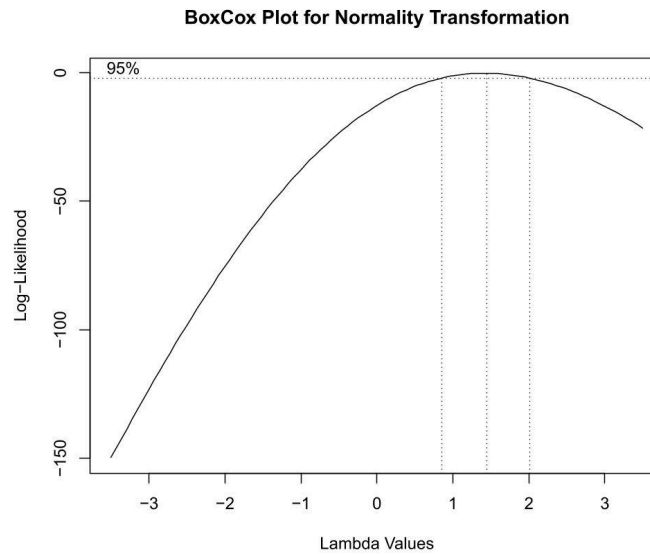


Figure 2: Box Cox transformation for normality

The Box-Cox plot displays the log-likelihood of different values for the transformation parameter λ . The maximum log-likelihood value corresponds to the best estimate for λ . The dotted lines on either side of the peak represent the 95% confidence interval for λ . From Figure 2, $\lambda = 1.4$ is an appropriate value for the transformation of the variable 'grad_rate' (y-value). It falls within the 95% confidence interval shown in the graph, which is approximately between 1.1 and 2. Thus, while transforming 'grad_rate' using $grad_rate_adj = grad_rate^{1.4}$ may not be physically meaningful, it is a valid method for achieving normality given the data. Since $\lambda = 1.4$ is within the 95% confidence interval, it is a suitable choice for the normality transformation.

To ensure that the normality transformation worked, I created a histogram and a normal QQ-plot of the transformed data. The histogram has a bell-shaped curve, indicating a normal distribution. The QQ-plot shows the data points a slightly improved straight line and improved tails, suggesting better normality. This confirms that the transformed response variable is close enough to normally distributed.

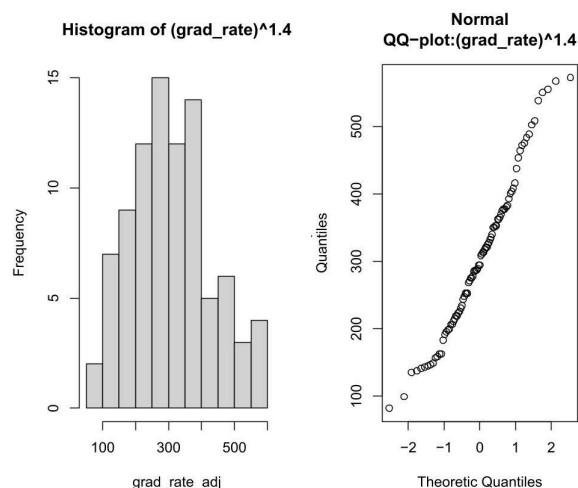


Figure 3: Normality checking of transformed data, $(\text{grad_rate})^{1.5}$

4 – X-Y Linearization

Now, we need to validate the x-y linearity relationships. To do this, we need to use scatter plots. This step allows us to evaluate how the regressors and response variables are related. If there is a non-linear relationship between any x-variables and the response variable, the x-variable needs to be transformed to create a linear relationship with the y-variable.

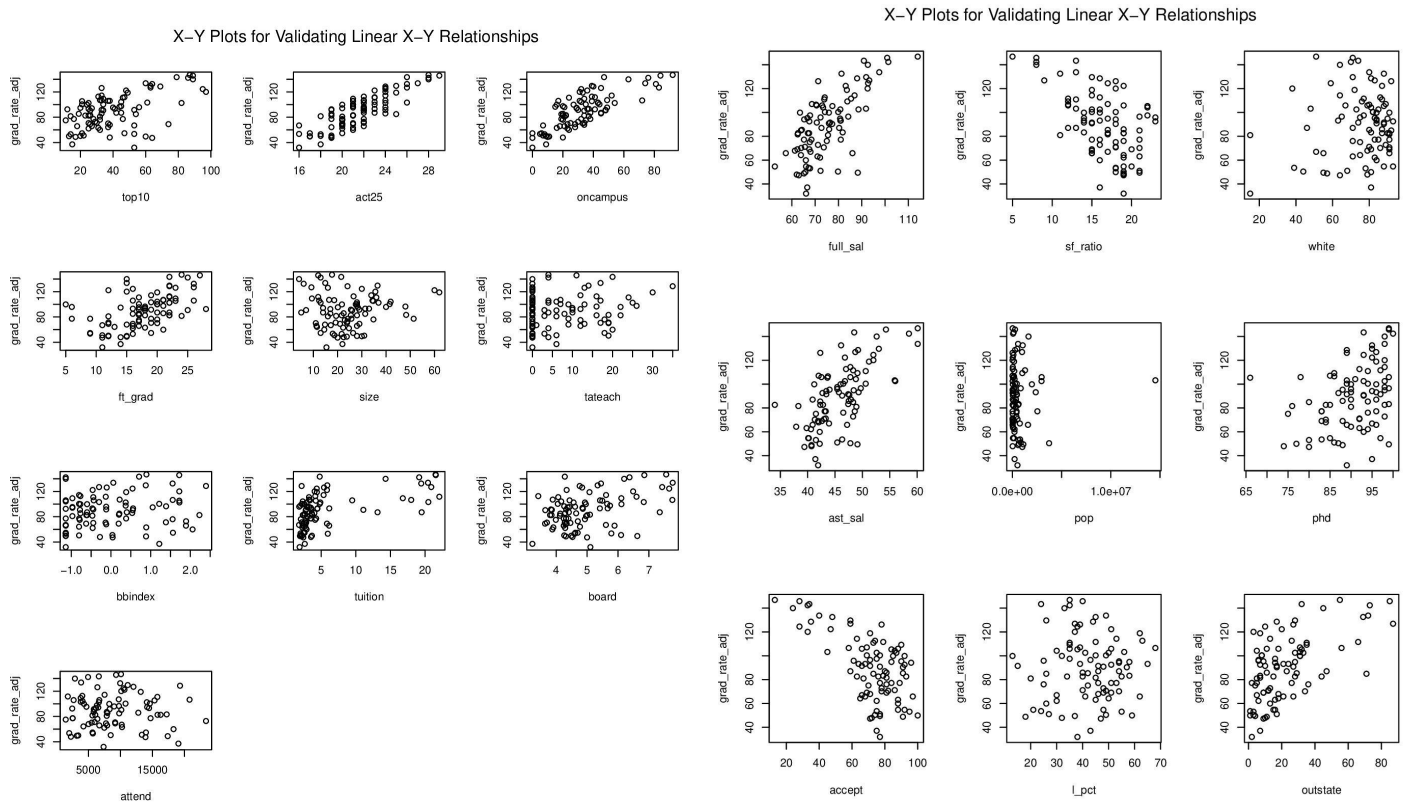
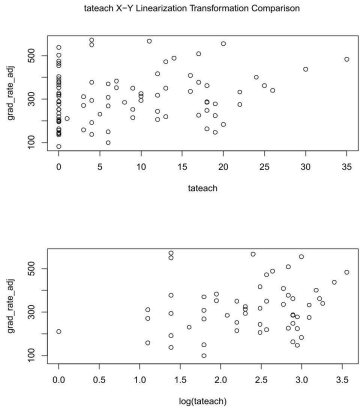
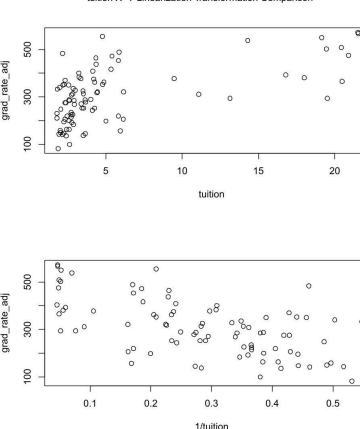
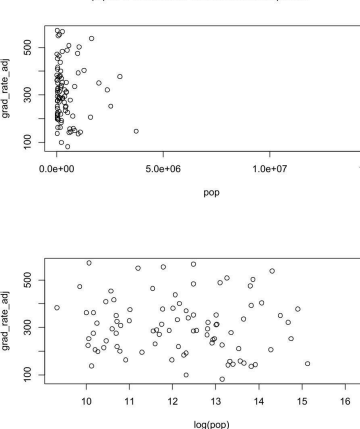
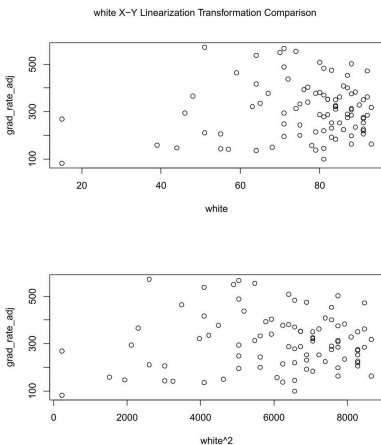
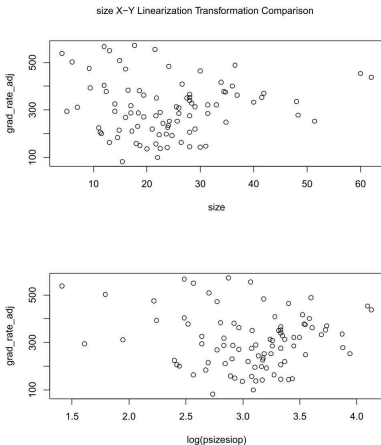


Figure 4: XY plots for validating XY linear relationships

After analyzing the above plots, it is clear to see that most of the x-variables are linearly related to **grade_rate_adj**. **Linearly aligned values: top10, act25, oncampus, ft_grad, l_pct, bbindex, board, attend, full_sal, ast_sal, sf_ratio, accept, phd, outstate.**

The variables that I chose to linearly transform are **tateach, tuition, pop, white and size**. We can see slight improvements in the linearity of these variables, although the improvement is *very slight*. However, some improvement is better than no improvement!

x-variable	analysis	plot	transformation
tateach	The log transformation has made the data points more spread out across the graph, rather than clustered at 0. Therefore, this transformation made tateach slightly more linear. However, there are now infinite data points, so this is not valid.		No transformation
tuition	The inverse transformation made the data points less clustered around 0-5 and more spread out with a downward trend, making the graph more linear.		Inverse transformation
pop	Although the log transformation didn't make the graph more linear, it did spread out the data points more, which is an improvement from the original graph.		Logarithmic transformation

white	The squared transformation made the data points less clustered around 80 and gave a wider spread to the data, creating a somewhat direct linear trend, an improvement from the original.		Squared transformation
size	The logarithmic transformation made the data points less clustered around the beginning and gave a better spread, showing an almost inverse relationship, an improvement from the original.		Logarithmic transformation

5 – VIF analysis

Variance Inflation Factor (VIF) screening is a statistical technique used to detect multicollinearity in multiple regression models. Multicollinearity occurs when two or more predictor variables in a model are highly correlated, which can inflate the variance of the coefficient estimates and make the model unreliable. Using R, we found the VIF values of each of the x-variables. If a VIF value is **greater than or equal to 5**, the x-variable could potentially have collinearity against other x-variables. By addressing multicollinearity, we can ensure that the model coefficients are more stable and that each predictor variable's contribution is more clearly understood. This leads to a more robust and interpretable model, providing better insights from the data.

By implementing the **vifstep** function in R, we can remove variables that have VIF values larger than 5 sequentially. Based on the screening, 3/19 variables had a collinearity problem: **full_sal**, **act25**, **oncampus**.

	Variables	VIF
1	top10	3.837657
2	act25	7.456859
3	oncampus	6.476489
4	ft_grad	2.315189
5	size	2.470994
6	tateach	1.488274
7	bhindex	3.143009
8	tuition	4.277794
9	board	2.545592
10	attend	2.687975
11	full_sal	8.233475
12	sf_ratio	3.346323
13	white	2.813517
14	ast_sal	5.732196
15	pop	1.888779
16	phd	1.850394
17	accept	5.515827
18	l_pct	1.686323
19	outstate	3.230831

Table 2: VIF analysis of x-variables

```

3 variables from the 19 input variables have collinearity problem:

full_sal act25 oncampus

After excluding the collinear variables, the linear correlation coefficients ranges
between:
min correlation ( attend ~ ft_grad ): -0.002631732
max correlation ( accept ~ top10 ): -0.7801415

----- VIFs of the remained variables -----
Variables      VIF
1      top10 3.374656
2      ft_grad 1.990453
3      size 2.143028
4      tateach 1.383878
5      bbindex 2.988605
6      tuition 3.200336
7      board 2.398218
8      attend 2.612676
9      sf_ratio 3.160678
10     white 2.206038
11     ast_sal 3.716321
12     pop 1.760122
13     phd 1.592489
14     accept 4.281381
15     l_pct 1.564903
16     outstate 2.939020

```

Table 3: VIF screening of high collinearity x-variables

Now that we have removed high collinearity x-variables, we can create a new multiple linear regression model using the transformed variable **grad_rate_adj** optimized through the Box-Cox transformation and the regressors that are left after the VIF screening.

```

Call:
lm(formula = grad_rate_adj ~ top10 + ft_grad + size + tateach +
    bbindex + tuition + board + attend + sf_ratio + white + ast_sal +
    pop + phd + accept + l_pct + outstate, data = exam2data_mod)

Residuals:
    Min       1Q   Median       3Q      Max
-119.301  -38.777   -2.393   41.348   93.921

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.340e+01  1.595e+02  -0.209  0.834703
top10        5.079e-01  5.538e-01   0.917  0.362114
ft_grad      5.993e+00  1.744e+00  3.436  0.000984 ***
size         3.627e+01  1.749e+01  2.073  0.041706 *
tateach      2.039e+00  8.237e-01  2.476  0.015648 *
bbindex      -1.515e+01  1.054e+01  -1.437  0.154937
tuition      -2.124e+02  7.496e+01  -2.833  0.005976 **
board         7.606e+00  8.735e+00  0.871  0.386733
attend       -3.969e-04  1.997e-03  -0.199  0.843021
sf_ratio     -2.214e+00  2.928e+00  -0.756  0.451886
white         1.534e-02  4.519e-03  3.394  0.001124 **
ast_sal      4.306e+00  2.451e+00  1.757  0.083168 .
pop          -3.527e+00  5.379e+00  -0.656  0.514120
phd           1.128e+00  1.085e+00  1.039  0.302080
accept       -2.080e+00  7.027e-01  -2.961  0.004154 **
l_pct        -1.342e+00  6.037e-01  -2.223  0.029378 *
outstate     4.539e-01  5.253e-01  0.864  0.390395

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 56.7 on 72 degrees of freedom
Multiple R-squared:  0.8086, Adjusted R-squared:  0.7661
F-statistic: 19.02 on 16 and 72 DF, p-value: < 2.2e-16

```

Figure 5: Regression Output after VIF Screening and adjusted y* data (grad_rate_adj)

6 – Stepwise Forward Selection

6.1 – Variable Selection

By using the stepwise forward selection, we start with an initial model and sequentially add **one regressor at a time**. We continuously check on the model performance and add variables until we reach the desired stopping point based on some performance measurement, in this case, alpha-to-enter (0.015).

To know the order of correlation for the x-variables, we need to know which variables are most correlated and start with that variable in the forward selection process. In R, a correlation matrix was created to find the correlation values of each x-variable. The variable with the most correlation was selected first to be the one-variable model, then so on.

If we order the correlations, the order is:

grad_rate_adj	ast_sal	accept	outstate	tuition	top10
1.0000000000	0.6574443660	-0.6467531831	0.6036748942	-0.5946490386	0.5554409720
sf_ratio	ft_grad	board	phd	bbindex	tateach
-0.5529543327	0.5056063770	0.4265151449	0.3707242187	0.1655331650	0.2012978432
white	size	pop	l_pct	attend	
0.0719725966	-0.0930452652	-0.0652256920	-0.0304693121	0.0004481661	

Because I ordered the variables based on their correlation, I wrote all R codes where I created printouts or slope models **in the order of the absolute value of correlation**.

After identifying that **ast_sal** has the highest correlation value, we can start to create our one variable model. The forward selection process is outlined in **Table 3**. There were 5 steps in the process. This is because once the p-value of the model is above the alpha-to-enter (0.015) we have to stop the process. The p-value for top10 (0.4301) is more than the alpha-to-enter (0.015), so the model ends at 4 variables: **ast_sal, accept, outstate, tuition**. See the [appendix \(page 26\)](#) for the partial F-statistic values and the p-values of each step.

Step #	# vars in model	Variable to enter	Partial F-Stat value	p-value	alpha-to-enter	p-value > alpha?
1	1	ast_sal	N/A	2.2e-16	0.015	N
2	2	accept	13.133	0.0004904	0.015	N
3	3	outstate	18.53	4.46e-5	0.015	N
4	4	tuition	9.54	0.002725	0.015	N
5	5	top10	0.6286	0.4301	0.015	Y

Table 3: Stepwise Forward Selection Process

6.2 – Model Fitting

Using the 4 variables identified in the variable selection process (**ast_sal, accept, outstate, tuition**) we can create a final stepwise forward selection model and continue to run our multiple linear regression.

```
Call:
lm(formula = grad_rate_adj ~ ast_sal + accept + outstate + tuition,
    data = newdata)

Residuals:
    Min       1Q   Median       3Q      Max
-150.99  -48.55   -9.82   56.94  169.63

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  151.4459   139.4985   1.086  0.28074
ast_sal       6.7420     2.2832   2.953  0.00408 **
accept      -1.7127     0.6066  -2.824  0.00593 **
outstate     1.3779     0.4921   2.800  0.00634 **
tuition    -203.6490    65.9340  -3.089  0.00273 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72.32 on 84 degrees of freedom
Multiple R-squared:  0.6368, Adjusted R-squared:  0.6196
F-statistic: 36.83 on 4 and 84 DF, p-value: < 2.2e-16
```

Figure 6: Model summary for model fitted using stepwise forward selection

According to the **Figure 6**, the new model should be:

$$y^* = 151.4469 + 6.7420X_{i,ast_sal} - 1.7127X_{i,accept} + 1.3779X_{i,outstate} - 203.6590X_{i,tuition}$$

The residual standard error is 72.32 on 84 degrees of freedom. The adjusted R-squared is 61.96%, the F-statistic is 36.83 and the p-value is 2.2e-16.

This model is significant as all of the variables have ** (p-values are under 0.01) and the adjusted R-squared is relatively high at 61.96%.

6.3 – Model Diagnostics

Using R, we can create residual diagnostic plots to see if the assumptions of the model hold. **Figure 7** shows the diagnostic plots.

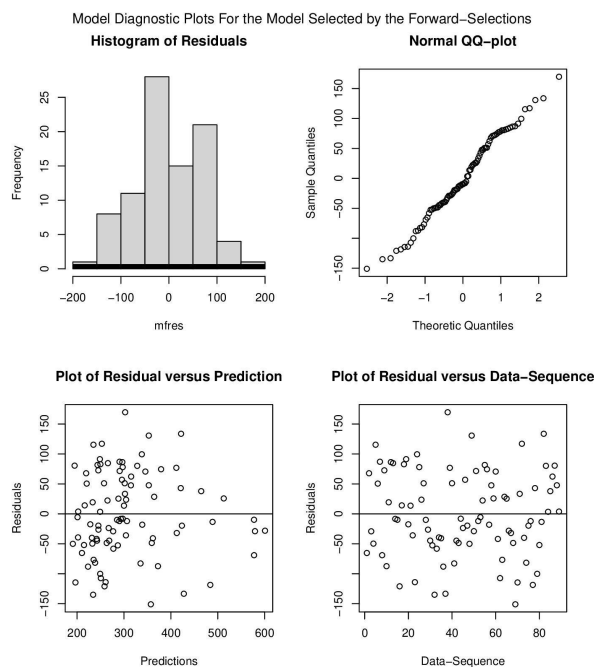


Figure 7: Model Diagnostic Plots for the Model Selected by the Forward Selections

We now check for if the assumptions hold as seen in **Figure 7**.

1. Errors are normally distributed

The Normal QQ-plot shows that the points are in approximately a straight line, **we can assume that the residuals are normally distributed.**

2. Errors have a mean of 0

The histogram of residuals show the residuals are symmetrical around 0. The residuals vs data-sequence plot shows that there is an equal distribution of residuals above/below 0. **The errors have a mean of 0.**

3. Errors have constant variance

The Residual vs Prediction plot shows an approximately equal spread of residuals across the horizontal axis. **We can assume that the errors have constant variance.**

4. Errors are independent

The residual vs Data-sequence plot shows no pattern amongst the data points. **The data points are independent from each other.**

5. Regressors are independent and identically distributed

We can use VIF to see if the regressor variables are correlated. If the variables have VIF values that are greater than 5, it indicates multicollinearity. There are no variables with VIF over 5 as seen in **Table 4. Therefore, we can assume that the regressors are independent.**

variable	VIF
ast_scl	3.71
accept	4.28
outstate	2.39
tuition	3.20

Table 4: VIF analysis for model fitted variables using stepwise forward selection

7 – All Subsets Regression

The all-subsets variable selection method involves all of the subset combinations of the regressor variables. They are then analyzed and the best subset is chosen based on a certain metric. In this exam, we are using both adjusted R^2 and the Cp-statistic.

7.1 - Adjusted R-Squared

First, we use the adjusted R^2 metric to determine the best model. Compared to a model with additional input variables, a higher adjusted R^2 indicates that the additional input variables are adding value to the model.

	Nvar	AdjR2	Cp	BIC
1	1	0.4257070	128.626302	-41.40067
2	1	0.4116034	133.872610	-39.24139
3	2	0.5970402	65.170639	-69.47390
4	2	0.5890197	68.119818	-67.71985
5	3	0.6590913	42.896472	-80.90899
6	3	0.6402812	49.732630	-76.12898
7	4	0.7231473	20.432870	-95.99711
8	4	0.6913784	31.842826	-86.32900
9	5	0.7333044	17.644615	-95.90096
10	5	0.7291224	19.128721	-94.51620
11	6	0.7462868	13.952636	-96.93254
12	6	0.7446965	14.510223	-96.37640
13	7	0.7588070	10.531767	-98.03996
14	7	0.7520719	12.864339	-95.58876
15	8	0.7654851	9.216242	-97.15591
16	8	0.7607529	10.834917	-95.37787
17	9	0.7671420	9.653892	-94.41781
18	9	0.7671119	9.664063	-94.40630
19	10	0.7717513	9.121087	-92.84230
20	10	0.7703769	9.579420	-92.30803
21	11	0.7741232	9.364283	-90.43178
22	11	0.7727919	9.802584	-89.90875
23	12	0.7737087	10.533197	-86.94339
24	12	0.7733770	10.640983	-86.81303
25	13	0.7726424	11.907593	-83.21519
26	13	0.7717623	12.189825	-82.87133
27	14	0.7709236	13.479300	-79.25092
28	14	0.7706268	13.573217	-79.13567
29	15	0.7691947	15.039500	-75.30398
30	15	0.7679438	15.429915	-74.82295

Figure 8: AdjR2, Cp-stat, BIC for regressor variables (*best adjR2 highlighted*)

After the R code performed the analysis needed to determine the best subsets of variables for each model size, we must now get the measurement metrics to determine the best model. Figure 8 provides the adjusted R^2 , Cp-statistic and BIC. I highlighted the model with the highest adjusted R^2 in red.

From **Figure 8**, the best model has 11 variables. Therefore, in the table of all of the variables for each model size, we will select the model with 11 variables. The variables that should be included are indicated with a *.

		top10	ft_grad	size	tateach	bbindx	board	tuition	attend	sf_ratio	white	ast_sal	pop	phd	accept	l_pct	outstate
1	(1)	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "
1	(2)	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "
2	(1)	" "	"*	" "	" "	" "	" "	" "	" "	" "	" "	"*	" "	" "	" "	" "	" "
2	(2)	" "	"*	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "	" "
3	(1)	" "	"*	" "	" "	" "	" "	" "	" "	" "	" "	"*	" "	" "	" "	" "	" "
3	(2)	" "	" "	" "	" "	" "	" "	" "	" "	" "	"*	"*	" "	" "	" "	" "	" "
4	(1)	" "	"*	" "	" "	" "	" "	" "	" "	" "	"*	"*	" "	" "	" "	" "	" "
4	(2)	" "	" "	" "	" "	" "	" "	"*	" "	" "	"*	"*	" "	" "	" "	" "	" "
5	(1)	" "	"*	" "	" "	" "	" "	"*	" "	" "	"*	"*	" "	" "	" "	" "	" "
5	(2)	" "	"*	" "	" "	" "	" "	" "	"*	"*	"*	"*	" "	" "	" "	" "	" "
6	(1)	" "	"*	" "	"*	" "	" "	"*	" "	" "	"*	"*	" "	" "	" "	" "	" "
6	(2)	" "	"*	" "	" "	" "	" "	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
7	(1)	" "	"*	" "	"*	" "	" "	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
7	(2)	"*	"*	" "	"*	" "	" "	"*	" "	" "	"*	"*	" "	" "	" "	" "	" "
8	(1)	" "	"*	" "	"*	"*	" "	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
8	(2)	" "	"*	" "	"*	" "	"*	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
9	(1)	" "	"*	"*	"*	"*	" "	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
9	(2)	" "	"*	" "	"*	"*	"*	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
10	(1)	" "	"*	"*	"*	"*	" "	"*	" "	" "	"*	"*	" "	" "	" "	"*	"*
10	(2)	" "	"*	"*	"*	"*	"*	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
11	(1)	" "	"*	"*	"*	"*	" "	"*	" "	" "	"*	"*	" "	" "	" "	"*	"*
11	(2)	" "	"*	"*	"*	"*	"*	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
12	(1)	" "	"*	"*	"*	"*	" "	"*	" "	" "	"*	"*	" "	" "	" "	"*	" "
12	(2)	" "	"*	"*	"*	"*	"*	"*	" "	" "	"*	"*	" "	" "	" "	"*	"*
13	(1)	"*	"*	"*	"*	"*	" "	"*	" "	" "	"*	"*	" "	" "	" "	"*	"*
13	(2)	"*	"*	"*	"*	"*	" "	"*	" "	" "	"*	"*	"*	" "	" "	"*	"*
14	(1)	"*	"*	"*	"*	"*	"*	"*	" "	"*	"*	"*	" "	" "	" "	"*	"*
14	(2)	"*	"*	"*	"*	"*	"*	"*	" "	" "	"*	"*	"*	"*	" "	"*	"*
15	(1)	"*	"*	"*	"*	"*	"*	"*	" "	"*	"*	"*	"*	"*	"*	"*	"*
15	(2)	"*	"*	"*	"*	"*	"*	"*	"*	"*	"*	"*	"*	"*	"*	"*	"*

Figure 9: Table with best subset of variables for each model size after comparisons in R

The highlighted model, which has 11 variables, has the highest adjusted R². The variables are: **ft_grad**, **size**, **tateach**, **bindex**, **tuition**, **white**, **ast_sal**, **phd**, **accept**, **l_pct**, **outstate**.

$$y = \beta_1(ft_grad) + \beta_2(size) + \beta_3(tateach) + \beta_4(bindex) + \beta_5(tuition) + \beta_6(white) + \beta_7(ast_sal) + \beta_8(phd) + \beta_9(l_pct) + \beta_{10}(outstate)$$

From the MLR using this equation done in R, the following printout is made. The variables are ordered in descending order of correlation:

```
Call:
lm(formula = grad_rate_adj ~ ft_grad + size + tateach + bindex +
    tuition + white + ast_sal + phd + accept + l_pct + outstate,
    data = exam2data_mod)

Residuals:
    Min       1Q   Median       3Q      Max
-111.216  -41.015    2.274   43.887  103.058

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -4.434e+01  1.359e+02  -0.326  0.745112
ft_grad      6.459e+00  1.593e+00   4.054  0.000120 ***
size        3.171e+01  1.628e+01   1.948  0.055066 .
tateach      1.920e+00  7.804e-01   2.460  0.016132 *
bindex      -1.940e+01  7.164e+00  -2.708  0.008335 **
tuition     -2.209e+02  6.490e+01  -3.403  0.001060 **
white        1.425e-02  3.766e-03   3.784  0.000304 ***
ast_sal      4.730e+00  2.243e+00   2.109  0.038218 *
phd          1.379e+00  1.023e+00   1.349  0.181377
accept      -2.685e+00  5.051e-01  -5.316  1e-06 ***
l_pct       -1.430e+00  5.690e-01  -2.512  0.014083 *
outstate     7.244e-01  4.574e-01   1.584  0.117397
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 55.72 on 77 degrees of freedom
Multiple R-squared:  0.8024,    Adjusted R-squared:  0.7741
F-statistic: 28.42 on 11 and 77 DF,  p-value: < 2.2e-16
```

Figure 10: Computer Printout from MLR of 11 variable model with adjusted R^2 as the metric

Figure 10 indicates that **ft_grad**, **tateach**, **bindex**, **tuition**, **white**, **ast_sal**, **accept**, **l_pct** are significant at $\alpha = 0.05$. The model is significant overall with an adjusted R-squared of 77.41% and a residual standard error of 55.72.

Next, using R, we need residual plots to check the assumptions that are associated with a MLR. Figure 11 shows the model diagnostic plots.

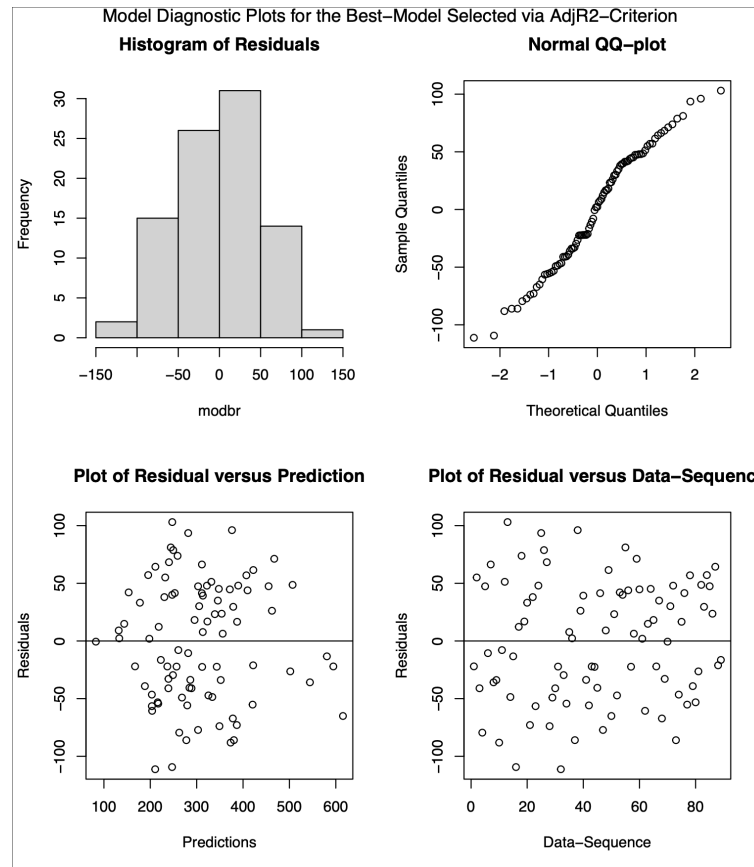


Figure 11: Model Diagnostic Plots for the Model Selected by Adjusted R-squared

1. Errors are normally distributed

The Normal QQ-plot shows that the points are in approximately a straight line, and the histogram is approximately bell shaped; we can **assume that the residuals are normally distributed.**

2. Errors have a mean of 0

The histogram of residuals show the residuals are symmetrical around 0. The residuals vs data-sequence plot shows that there is an equal distribution of residuals above/below 0. **The errors have a mean of 0.**

3. Errors have constant variance

The Residual vs Prediction plot shows an approximately equal spread of residuals across the horizontal axis. **We can assume that the errors have constant variance.**

4. Errors are independent

The residual vs Data-sequence plot shows no pattern amongst the data points. **The data points are independent from each other.**

5. Regressors are independent identically distributed

We can use VIF to see if the regressor variables are correlated. Some variables have VIF values that are greater than 5, indicating multicollinearity as seen in **Table 5. Therefore, we cannot assume that the regressors are independent.**

variable	VIF	variable	VIF
ft_grad	2.315189	ast_sal	5.732196
size	2.470994	phd	1.850394
tateach	1.488274	accept	5.515827
bbindex	3.143009	l_pct	1.686323
tuition	4.277794	outstate	3.230831
white	2.545592		

Table 5: VIF values of Regressor Variables in 11 Variable Model

7.2 - Cp-Statistic

The Cp-Statistic helps to balance the trade-off between model fit and complexity, guiding the selection of models that provide a good approximation to the full model without including unnecessary predictors. By choosing models with Cp values close to and greater than the number of predictors, you can ensure that the selected model is likely to generalize well to new data.

	Nvar	AdjR2	Cp	BIC
1	1	0.4257070	128.626302	-41.40067
2	1	0.4116034	133.872610	-39.24139
3	2	0.5970402	65.170639	-69.47390
4	2	0.5890197	68.119818	-67.71985
5	3	0.6590913	42.896472	-80.90899
6	3	0.6402812	49.732630	-76.12898
7	4	0.7231473	20.432870	-95.99711
8	4	0.6913784	31.842826	-86.32900
9	5	0.7333044	17.644615	-95.90096
10	5	0.7291224	19.128721	-94.51620
11	6	0.7462868	13.952636	-96.93254
12	6	0.7446965	14.510223	-96.37640
13	7	0.7588070	10.531767	-98.03996
14	7	0.7520719	12.864339	-95.58876
15	8	0.7654851	9.216242	-97.15591
16	8	0.7607529	10.834917	-95.37787
17	9	0.7671420	9.653892	-94.41781
18	9	0.7671119	9.664063	-94.40630
19	10	0.7717513	9.121087	-92.84230
20	10	0.7703769	9.579420	-92.30803
21	11	0.7741232	9.364283	-90.43178
22	11	0.7727919	9.802584	-89.90875
23	12	0.7737087	10.533197	-86.94339
24	12	0.7733770	10.640983	-86.81303
25	13	0.7726424	11.907593	-83.21519
26	13	0.7717623	12.189825	-82.87133
27	14	0.7709236	13.479300	-79.25092
28	14	0.7706268	13.573217	-79.13567
29	15	0.7691947	15.039500	-75.30398
30	15	0.7679438	15.429915	-74.82295

Figure 12: AdjR2, Cp-stat, BIC for regressor variables (best Cp-statistic highlighted)

Based on **Figure 9**, the model with the best Cp-Statistic has 15 variables. The Cp-Statistic is 15.039500, which is the closest and greater than to the number of variables of the model, in this case, 15. The model is highlighted in red.

The highlighted model has the variables: **top10**, **ft_grad**, **size**, **tateach**, **bindex**, **board**, **tuition**, **sf_ratio**, **white**, **ast_sal**, **pop**, **phd**, **accept**, **l_pct**, **outstate** as seen in Figure 9.

$$y^* = \beta_1(\text{top10}) + \beta_2(\text{ft_grad}) + \beta_3(\text{size}) + \beta_4(\text{tateach}) + \beta_5(\text{bindex}) + \beta_6(\text{board}) + \beta_7(\text{tuition}) + \beta_8(\text{sf_ratio}) + \beta_9(\text{white}) + \beta_{10}(\text{ast_sal}) + \beta_{11}(\text{pop}) + \beta_{12}(\text{phd}) + \beta_{13}(\text{accept}) + \beta_{14}(\text{l_pct}) + \beta_{15}(\text{outstate})$$

From the MLR in R, we get the following printout:

```
Call:
lm(formula = grad_rate_adj ~ top10 + ft_grad + size + tateach +
    bindex + board + tuition + sf_ratio + white + ast_sal +
    pop + phd + accept + l_pct + outstate, data = exam2data_mod)

Residuals:
    Min       1Q   Median       3Q      Max
-117.598  -35.710   -3.373   40.521   96.017

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.487e+01  1.582e+02  -0.220  0.826206
top10         5.062e-01  5.501e-01   0.920  0.360487
ft_grad       6.005e+00  1.731e+00   3.468  0.000882 ***
size         3.576e+01  1.719e+01   2.080  0.041021 *
tateach       2.059e+00  8.121e-01   2.536  0.013358 *
bindex      -1.661e+01  7.531e+00  -2.205  0.030597 *
board        7.540e+00  8.671e+00   0.870  0.387362
tuition     -2.162e+02  7.198e+01  -3.004  0.003654 **
sf_ratio    -2.037e+00  2.770e+00  -0.735  0.464444
white        1.521e-02  4.445e-03   3.422  0.001022 **
ast_sal      4.314e+00  2.434e+00   1.772  0.080546 .
pop         -3.565e+00  5.340e+00  -0.668  0.506506
phd          1.122e+00  1.078e+00   1.041  0.301377
accept      -2.087e+00  6.973e-01  -2.993  0.003763 **
l_pct       -1.356e+00  5.955e-01  -2.277  0.025718 *
outstate     4.658e-01  5.184e-01   0.899  0.371841

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 56.33 on 73 degrees of freedom
Multiple R-squared:  0.8085,    Adjusted R-squared:  0.7692
F-statistic: 20.55 on 15 and 73 DF,  p-value: < 2.2e-16
```

Figure 13. Computer Printout from MLR of 15 variable model using Cp-Statistic as Metric

This model shows that the variables **ft_grad**, **size**, **tateach**, **bindex**, **tuition**, **white**, **accept**, **l_pct** are significant with an $\alpha = 0.05$, and the model is significant overall with an adjusted R-squared of 76.92% and a residual standard error of 56.33.

Next, we use R to create residual diagnostic plots to see if the assumptions of normality hold.

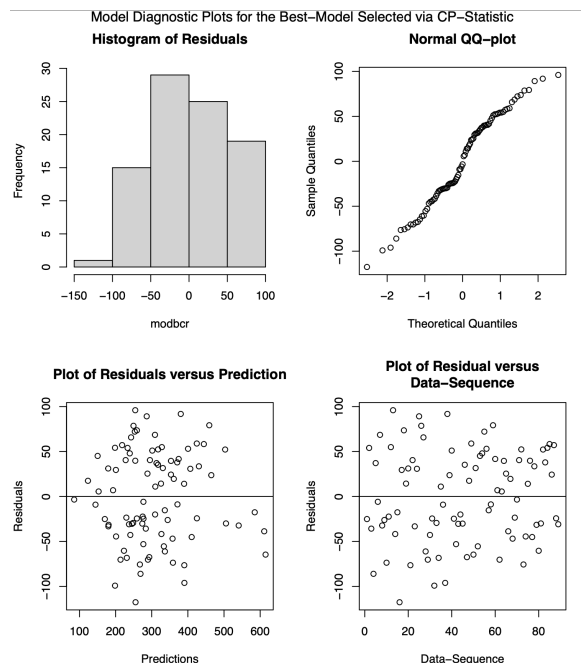


Figure 14. Model Diagnostic Plots for the Model Selected by the Cp-Statistic

1. Errors are normally distributed

The Normal QQ-plot shows that the points are in approximately a straight line, but the histogram is left-skewed; **we cannot assume that the residuals are normally distributed.**

2. Errors have a mean of 0

The histogram of residuals show the residuals are symmetrical around 0. The residuals vs data-sequence plot shows that there is an equal distribution of residuals above/below 0. **The errors have a mean of 0.**

3. Errors have constant variance

The Residual vs Prediction plot shows an approximately equal spread of residuals across the horizontal axis. **We can assume that the errors have constant variance.**

4. Errors are independent

The residual vs Data-sequence plot shows no pattern amongst the data points. **The data points are independent from each other.**

5. Regressors are independent identically distributed

We can use VIF to see if the regressor variables are correlated. Some variables have VIF values that are greater than 5, indicating multicollinearity as seen in **Table 6. Therefore, we cannot assume that the regressors are independent.**

variable	VIF	variable	VIF
top10	3.837657	white	2.813517
ft_grad	2.315189	ast_sal	5.732196
size	2.470994	pop	1.888779
tateach	1.488274	phd	1.850394
bbindex	3.143009	accept	5.515827
board	2.545592	l_pct	1.686323
tuition	4.277794	outstate	3.230831
sf_ratio	3.346323		

Table 6: VIF values of Regressor Variables in 15 Variable Model

7.3 – Comparison Between Adjusted R-Squared and Cp-Statistic Model

The model selected from the best adjusted R-squared had 11 variables and the model selected from the best Cp-statistic had 15 variables. After running MLR for both models, the adjusted R-squared model performed better. The adjusted R-squared model had a better standard residual error than the Cp-Statistic model (55.72 vs 56.33). Both models had the same amount of significant variables at $\alpha = 0.05$, but because the residual standard error is better for the adjusted R-squared model, I am inclined to believe that adjusted R-squared is a better metric for whether or not a model is best or not. It is also a simpler model to work with.

8 – Conclusion and Answering the Question

Do successful sports programs raise graduation rates?

The analysis of NCAA graduation rates demonstrates that successful sports programs positively impact graduation rates. The study utilized multiple linear regression (MLR) analysis on data from NCAA Division I universities, considering variables such as the composite basketball ranking (bbindex) and average basketball home attendance (attend). These variables were significant predictors in models that showed improved graduation rates. By incorporating these sports-related factors, the adjusted R-squared model indicated a better fit and explained more variance in graduation rates compared to models excluding these variables. This suggests that robust intercollegiate athletics programs contribute to higher academic success and graduation rates among students.

What have I learned from this exam?

After conducting multiple linear regression analyses using various variable selection methods, I observed that the all-subset selection approach produced a more robust and accurate model compared to others. This aligns with my coursework, where forward stepwise selection is known for its efficiency in generating simpler models, while all-subset selection is recognized for its thoroughness and superior performance. Engaging with a real-world dataset allowed me to apply multiple linear regression techniques comprehensively. This experience has enhanced my confidence in applying these regression methods to practical, real-life situations.

Appendix: All R Codes Used in the Data Analyzation Process

Data removed: 82, 55, 91, 22, 72

R Codes

#1.1 – Setting and getting the directory

```
setwd("/Users/nityapatil/Desktop/ISyE4031/computer projects/EXAM 2")
getwd()
```

#1.2 – Installing “leaps” and “usdm” packages / loading them to library

```
#installing "leaps" and usdm"
install.packages("leaps")
library("leaps")
install.packages("usdm")
library("usdm")
```

#1.3 – Loading data into R and outputting data

```
exam2data = read.table("2.1.1) NCAA-Grad-Rate.txt")
exam2data
```

#1.4 – Defining the y-variable

```
#defining y-variable: graduation rate
grad_rate = exam2data[,21]
grad_rate
```

#1.5 - Defining x-variables

```
top10 = exam2data[,2]
act25 = exam2data[,3]
oncampus= exam2data[,4]
ft_grad=exam2data[,5]
size=exam2data[,6]
tateach=exam2data[,7]
bbindex=exam2data[,8]
tuition=exam2data[,9]
board=exam2data[,10]
attend=exam2data[,11]
full_sal=exam2data[,12]
sf_ratio=exam2data[,13]
white=exam2data[,14]
ast_sal=exam2data[,15]
pop=exam2data[,16]
phd=exam2data[,17]
accept=exam2data[,18]
l_pct=exam2data[,19]
outstate=exam2data[,20]
```

#2 - Normal Distribution Evaluation of Y-Data

```
#creating a histogram and normal-QQ plot for y-data
attach(mtcars) #special R-adds-on package for combined plots
#external file to save
pdf(file = 'Normality Check For Response Variable.pdf', height = 6)
par(mfrow=c(1,2)) #1 by 2 plot
hist(grad_rate, main="Histogram of Graduation Rate")
qqnorm(grad_rate, main="Normal QQ-plot",ylab="Sample
Quantiles",xlab="Theoretical Quantiles")
dev.off()
```

#3.1 - Box-Cox Transformation of Response Variable

```
library("MASS") #load the library("mass") for using the BoxCox transformation
#running first path main-effect model
mod1 = lm(formula = grad_rate~ top10 + act25+ oncampus + ft_grad + size + tateach + bbindex + tuition +
board + attend + full_sal + sf_ratio + white + ast_sal + pop + phd + accept + l_pct + outstate,
exam2data)
pdf(file= 'Box-Cox Transformation of the Response Variable.pdf', height = 6)
par(mfrow=c(1,1)) #make it to one plot per sheet
boxcox(mod1, lambda = seq(-3.5, 3.5, 1/10), plotit = TRUE,
xlab = "Lambda Values", ylab = "Log-Likelihood")
title("BoxCox Plot for Normality Transformation")
#Turn Off the device for saving the R-plot in an external file.
dev.off()
```

#3.2 - Y-Variable Transformation

```
#adjusting response variable
grad_rate_adj = grad_rate ^ 1.4
```

#3.3 - Histogram and Normal-QQ Plot to Verify Transformation Normality

```
#external file to save the R-plot
pdf(file= 'Histogram and QQ-Norm of TRANSFORMED Response Variable.pdf', height = 6)
#checking histogram and QQ-normal plots for normality
par(mfrow=c(1,2))
```

```
hist(grad_rate_adj, main="Histogram of (grad_rate)^1.4")
qqnorm(grad_rate_adj, main="Normal
QQ-plot:(grad_rate)^1.4", ylab="Sample Quantiles",xlab="Theoretical Quantiles")
dev.off()
```

#4.1 - X-Y Linearization

```
#external file to save the plot
pdf(file= '[1] Y-X Linear Relationship Plots.pdf', height = 8)
#combine plot
par(mfrow=c(4,3)) #it will be a 4 by 3 plot
plot(top10, grad_rate_adj, xlab = "top10", ylab = "grad_rate_adj")
plot(act25, grad_rate_adj, xlab = "act25", ylab = "grad_rate_adj")
plot(oncampus, grad_rate_adj, xlab = "oncampus", ylab = "grad_rate_adj")
plot(ft_grad, grad_rate_adj, xlab = "ft_grad", ylab = "grad_rate_adj")
plot(size, grad_rate_adj, xlab = "size", ylab = "grad_rate_adj")
plot(tateach, grad_rate_adj, xlab = "tateach", ylab = "grad_rate_adj")
plot(bbindex, grad_rate_adj, xlab = "bbindex", ylab = "grad_rate_adj")
plot(tuition, grad_rate_adj, xlab = "tuition", ylab = "grad_rate_adj")
plot(board, grad_rate_adj, xlab = "board", ylab = "grad_rate_adj")
plot(attend, grad_rate_adj, xlab = "attend", ylab = "grad_rate_adj")
mtext("X-Y Plots for Validating Linear X-Y Relationships", side = 3, line = -2, outer = TRUE)
#Turn Off the device for saving the R-plot in an external file.
dev.off()
#external file to save the plot
pdf(file= '[2] Y-X Linear Relationship Plots.pdf',height = 8)
# --- let us use the combine-plot to organize the plots into a 5 rows by 4 columns plot
par(mfrow=c(3,3)) #3 by 3 plot
plot(full_sal,grad_rate_adj,xlab="full_sal",ylab="grad_rate_adj")
plot(sf_ratio,grad_rate_adj,xlab="sf_ratio",ylab="grad_rate_adj")
plot(white,grad_rate_adj,xlab="white",ylab="grad_rate_adj")
plot(ast_sal,grad_rate_adj,xlab="ast_sal",ylab="grad_rate_adj")
plot(pop,grad_rate_adj,xlab="pop",ylab="grad_rate_adj")
plot(phd,grad_rate_adj,xlab="phd",ylab="grad_rate_adj")
plot(accept,grad_rate_adj,xlab="accept",ylab="grad_rate_adj")
plot(l_pct,grad_rate_adj,xlab="l_pct",ylab="grad_rate_adj")
plot(outstate,grad_rate_adj,xlab="outstate",ylab="grad_rate_adj")
mtet("X-Y Plots for Validating Linear X-Y Relationships", side = 3, line = -2, outer = TRUE)
#Turn Off the device for saving the R-plot in an external file.
dev.off()
```

#4.2 - Transforming the non-linear x-variables

```
#tateach
pdf(file= 'tateach linearization.pdf', height = 8)
par(mfrow=c(2,1))
plot(tateach,grad_rate_adj,xlab="tateach",ylab="grad_rate_adj")
plot(log(tateach), grad_rate_adj,xlab="log(tateach)",ylab="grad_rate_adj")
mtext("tateach X-Y Linearization Transformation Comparison", side = 3, line = -2, outer = TRUE)
dev.off()
#tuition:
pdf(file= 'tuition linearization.pdf', height = 8)
par(mfrow=c(2,1))
plot(tuition,grad_rate_adj,xlab="tuition",ylab="grad_rate_adj")
plot(1/tuition, grad_rate_adj,xlab="1/tuition",ylab="grad_rate_adj")
mtext("tuition X-Y Linearization Transformation Comparison", side = 3, line = -2, outer = TRUE)
dev.off()
#pop
```

```
pdf(file= 'pop linearization.pdf', height = 8)
par(mfrow=c(2,1))
plot(pop,grad_rate_adj,xlab="pop",ylab="grad_rate_adj")
plot(log(pop), grad_rate_adj,xlab="log(pop)",ylab="grad_rate_adj")
mtext("pop X-Y Linearization Transformation Comparison", side = 3, line = -2, outer = TRUE)
dev.off()
#white
pdf(file= 'white linearization.pdf', height = 8)
par(mfrow=c(2,1))
plot(white,grad_rate_adj,xlab="white",ylab="grad_rate_adj")
plot(white^2, grad_rate_adj,xlab="white^2",ylab="grad_rate_adj")
mtext("white X-Y Linearization Transformation Comparison", side = 3, line = -2, outer = TRUE)
dev.off()
#size
pdf(file= 'size linearization.pdf', height = 8)
par(mfrow=c(2,1))
plot(size,grad_rate_adj,xlab="size",ylab="grad_rate_adj")
plot(log(size), grad_rate_adj,xlab="log(size)",ylab="grad_rate_adj")
mtext("size X-Y Linearization Transformation Comparison", side = 3, line = -2, outer = TRUE)
dev.off()
```

#4.3 - Non-linear x-variable adjustments

```
tuition = 1/tuition
pop = log(pop)
white = white^2
size = log(size)
```

#5.1 - Creating a new data set

```
#creating new data set
exam2data_ext = cbind(grad_rate_adj, top10, act25, oncampus, ft_grad, size, tateach, bbindex, tuition,
board, attend, full_sal, sf_ratio, white, ast_sal, pop, phd, accept, l_pct, outstate)
exam2data_ext = as.data.frame(exam2data_ext)
exam2data_ext
```

#5.2 - Reviewing VIF values for x-variables

```
#reviewing VIF values for x-variables
exam2data_x = exam2data_ext[,2:ncol(exam2data_ext)] #remove the Y data from the x-data matrix
vf = vif(exam2data_x)
sink("X-Variable VIF Values.txt") #vif values to external file
print(vf)
sink()
```

#5.3 - VIF step to remove variables with VIF values larger than 5

```
#VIF step function to remove variables with VIF values >= 5 sequentially
v_remain = vifstep(exam2data_x, th=5)
v_remain #print out the remaining x-variables
sink("VIF-Based Variable-Screenings.txt")
print(v_remain)
sink()
```

#5.4 - MLR model on adjusted y-variable and VIF selected x-variables

```
#new data matrix to include the y* and the VIF-screen X-data
exam2data_mod = cbind(grad_rate_adj, top10, ft_grad, size, tateach, bbindex, tuition, board, attend,
sf_ratio, white, ast_sal, pop, phd, accept, l_pct, outstate)
exam2data_mod = as.data.frame(exam2data_mod)
#regression model with variables screened by VIF
```

```

mod_1 = lm(formula = grad_rate_adj ~ top10 + ft_grad + size + tateach + bbindex + tuition + board +
attend + sf_ratio + white + ast_sal + pop + phd + accept + l_pct + outstate, exam2data_mod)
mod_1_summary = summary(mod_1)
mod_1_summary
#save regression summary
sink("Regressions with All X-Variables Screened by VIF.txt")
print(mod_1_summary)
sink()

```

#6.1 - Correlation Matrix

```

#start with x-variable most correlated to y*
cor_yx = cor(exam2data_mod) #this is correlation matrix for all x-y data-columns
cor_yax = cor_yx[,1] #Only the correlations between Y-data and other X-variables are needed.
cor_yax #print to see y-x correlations
#--- from the y-correlation column one can see which x-variable is most correlated (check the absolute
value of the correlation)
#--- Let us rearrange the X-data columns according to their abs(correlation) against Y.
newdata = cbind(grad_rate_adj, ast_sal, accept, outstate, tuition, top10, sf_ratio, ft_grad, board,
phd, bbindex, tateach, white, size, pop, l_pct, attend)
newdata
exam2data_mod = as.data.frame(exam2data_mod) #regression model with variables screened by VIF
mod_1 = lm(formula = grad_rate_adj ~ top10 + ft_grad + size + tateach + bbindex + board + tuition +
attend + sf_ratio + white + ast_sal + pop + phd + accept + l_pct + outstate, exam2data_mod)
mod_1_summary = summary(mod_1)
mod_1_summary
#save regression summary
sink("Regressions with All X-variables Screened by the VIF.txt")
print(mod_1_summary)
sink()

```

```

newdata = cbind(grad_rate_adj, ast_sal, accept, outstate, tuition, top10, sf_ratio, ft_grad, board,
phd, bbindex, tateach, white,
size, pop, l_pct, attend)
newdata = as.data.frame(newdata)
#--- Let us check out the correlations for this new data set.
cor_yx = cor(newdata) #this is correlation matrix for all x-y data-columns
cor_yax = cor_yx[,1] #Only the correlations between Y-data and other X-variables are needed.
cor_yax #print to see y-x correlations
#--- Keep the above Y-X correlation values to an external file----
sink("Y-X Correlation Values.txt")
print(cor_yax)
sink()

```

#6.2 - Stepwise Forward Selection Models

```

#model 1
model_one = lm(grad_rate_adj ~ ast_sal, newdata)
model_one_s = summary(model_one)
model_one_s
sink("One-variable model.txt")
print(model_one)
sink()
#model 2
reduced = model_one
model_two = lm(formula = grad_rate_adj ~ ast_sal + accept, newdata)
model_two_s = summary(model_two)
model_two_s

```

```

pf1 = anova(reduced, model_two)
pf1
sink("1st Partial F-Test.txt")
print(pf1)
sink()
#model 3
reduced = model_two
model_3 = lm(formula = grad_rate_adj ~ ast_sal + accept + outstate, newdata)
Pf2 = anova(reduced, model_3)
Pf2
sink("2nd Partial F-Test.txt")
print(Pf2)
sink()

#model 4
reduced = model_3
model_4 = lm(formula = grad_rate_adj ~ ast_sal + accept + outstate + tuition, newdata)
Pf3 = anova(reduced, model_4)
Pf3
sink("3rd Partial F-Test.txt")
print(Pf3)
sink()

#model 5
reduced = model_4
model_5 = lm(formula = grad_rate_adj ~ ast_sal + accept + outstate + tuition + top10, newdata)
Pf4 = anova(reduced, model_5)
Pf4
sink("4th Partial F-Test.txt")
print(Pf4)
sink()

#Final summary
mod_forward = lm(formula = grad_rate_adj ~ ast_sal + accept + outstate + tuition, newdata)
mod_forward_s = summary(mod_forward)
mod_forward_s
sink("Stepwise Forward Selection Model.txt")

#6.3 - Model Diagnostics
#the model residual = prediction-errors and also the predictions
mfres = mod_forward$residuals
mfpred = predict(mod_forward)
#external file
pdf(file= 'Regression Diagnostics - Forward-Selection Model.pdf', height = 8)
par(mfrow=c(2,2)) #it will be a 2 by 2 plot
hist(mfres, main="Histogram of Residuals")
qqnorm(mfres, main="Normal QQ-plot",ylab="Sample Quantiles",xlab="Theoretical Quantiles")
plot(mfpred, mfres, main="Plot of Residual versus Prediction", xlab="Predictions", ylab="Residuals")
abline(h = mean(mfres)) #place an horizontal line on expected-residual=0
nmf = length(exam2data_ext[,2]) #get the size of data to prepare the next plot
plot(c(1:nmf),mfres, main="Plot of Residual versus Data-Sequence", xlab="Data-Sequence",
ylab="Residuals")
abline(h = mean(mfres))
mtext("Model Diagnostic Plots For the Model Selected by the Forward-Selections", side = 3, line = -1,
outer = TRUE)
dev.off()

```


#7 - All-subsets Selection

```
moda = regsubsets(grad_rate_adj ~ top10 + ft_grad + size + tateach + bbindex + board + tuition + attend
+ sf_ratio + white + ast_sal + pop + phd + accept + l_pct + outstate,
data=exam2data_mod, nbest=2, nvmax = 15)
modas = summary(modas)
modas #print the results on the screen
sink("Best-Subset Regressions.txt")
print(modas)
sink()
#--- Let us get the Adjusted-R^2, Cp and BIC for each model calculated ---
c1 = c(1,1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,11,11,12,12,13,13,14,14,15) #Create one column of
no.-variables included in the model
modas_M = cbind(c1,modas$adjr2,modas$cp,modas$bic) #bind
various summary metrics into a data-matrix
modas_Mf = as.data.frame(modas_M) #need to change the data matrix into a R-data-frame
names(modas_Mf)[1] = "Nvar" #define names for the data columns in the data-frame object
names(modas_Mf)[2] = "AdjR2"
names(modas_Mf)[3] = "Cp"
names(modas_Mf)[4] = "BIC"
modas_Mf

#--- Keep the above best-subset regression's metrics to an external file ----
sink("Best-Subset Regressions' Metric Values.txt")
print(modas_Mf)
sink()
#--- Graphical presentations of the above results ---
#Let us prepare an external file to save the following R-plot.
pdf(file= 'Best-Subset Selection Metric Values.pdf', height = 8)
par(mfrow=c(2,2)) #it will be a 2 by 2 plot
plot(modas$adjr2,xlab="no. variables",ylab="Adjusted R^2",ylim=c(0.40,0.80))
plot(modas$cp,xlab="no. variables",ylab="Cp-value")
plot(modas$bic,xlab="no. variables",ylab="BIC") #no AIC, but BIC values is available
mtext("Best-Subsets Variable-Selection Plots", side = 3, line = -2, outer = TRUE)
dev.off()

#--- Graphical presentations of the above results ---
#Let us prepare an external file to save the following R-plot.
pdf(file= 'Best-Subset Selection Metric Values.pdf', height = 8)
par(mfrow=c(2,2)) #it will be a 2 by 2 plot
plot(modas$adjr2,xlab="no. variables",ylab="Adjusted R^2",ylim=c(0.40,0.80))
plot(modas$cp,xlab="no. variables",ylab="Cp-value")
plot(modas$bic,xlab="no. variables",ylab="BIC") #no AIC, but BIC values is available
mtext("Best-Subsets Variable-Selection Plots", side = 3, line= -2, outer = TRUE)
dev.off()

#Adjusted R-Squared
#--- Then, run a regular regression for the selected model ---
modb = lm(formula = grad_rate_adj ~ ft_grad + size + tateach + bbindex + tuition + white + ast_sal +
phd + accept + l_pct + outstate, exam2data_mod)
modb_s = summary(modb)
modb_s
#--- Keep the above model picked from the best-subset selections to an external file ----
sink("Regression Model(Best-Subset Selection Adjusted R^2).txt")
print(modb_s)
sink()
```

```

#--- One shall examine the model-diagnostic plots to make sure that all model assumptions are valid.
-----
#--- Let us get the model residual = prediction-errors and also the predictions first ---
modbr = modb$residuals
modbp = predict(modb)
#Let us prepare an external file to save the following R-plot.
pdf(file= 'Regression Diagnostics for the Best-Subset-Selected Model - adjusted R^2.pdf', height = 8)
par(mfrow=c(2,2)) #it will be a 2 by 2 plot
hist(modbr, breaks=6, main="Histogram of Residuals")
qqnorm(modbr, main="Normal QQ-plot",ylab="Sample Quantiles",xlab="Theoretical Quantiles")
plot(modbp, modbr, main="Plot of Residual versus Prediction",xlab="Predictions",ylab="Residuals")
abline(h = mean(modbr)) #place an horizontal line on expected-residual=0
nmf = length(exam2data_mod[,2]) #get the size of data to prepare the next plot
plot(c(1:nmf),modbr, main="Plot of Residual versus
Data-Sequence",xlab="Data-Sequence",ylab="Residuals")
abline(h = mean(modbr))
mtext("Model Diagnostic Plots for the Best-Model Selected via AdjR2-Criterion", side = 3, line = -1,
outer = TRUE)
#Turn Off the device for saving the R-plot in an external file.
dev.off()

```

7.2 - Cp-Statistic Analysis

```

#--- Then, run a regular regression for the selected model -
modbc = lm(formula = grad_rate_adj ~ top10 + ft_grad + size + tateach + bbindex + board + tuition +
sf_ratio + white + ast_sal + pop + phd + accept + l_pct + outstate, exam2data_mod)
modbc_s = summary(modbc)
modbc_s
#--- Keep the above model picked from the best-subset selections to an external file ----
sink("Regression Model (Best-Subset Selections CP-Statistic).txt")
print(modbc_s)
sink()
#--- Let us get the model residual = prediction-errors and also the predictions first ---
modbcr = modbc$residuals
modbcp = predict(modbc)
#Let us prepare an external file to save the following R-plot.
pdf(file= 'Regression Diagnostics for the Best-Subset-Selected Model - CP-Statistic.pdf', height = 8)
par(mfrow=c(2,2)) #it will be a 2 by 2 plot
hist(modbcr, breaks = 7, main="Histogram of Residuals")
qqnorm(modbcr, main="Normal QQ-plot",ylab="Sample Quantiles",xlab="Theoretical Quantiles")
plot(modbcp, modbcr, main="Plot of Residuals versus Prediction",xlab="Predictions",ylab="Residuals")
abline(h = mean(modbcr)) #place an horizontal line on expected-residual=0
nmf = length(exam2data_mod[,2]) #get the size of data to prepare the next plot
plot(c(1:nmf),modbcr, main="Plot of Residual versus
Data-Sequence",xlab="Data-Sequence",ylab="Residuals")
abline(h = mean(modbcr))
mtext("Model Diagnostic Plots for the Best-Model Selected via CP-Statistic", side = 3, line = -1, outer
= TRUE)
#Turn Off the device for saving the R-plot in an external file.
dev.off()

```

#Save History

```

save.image(file= "Exam2.rda")
savehistory(file = "Exam2.Rhistory")

```

Stepwise Forward Selection Printouts

First Partial F-Test

Analysis of Variance Table

Model 1: grad_rate_adj ~ ast_sal

Model 2: grad_rate_adj ~ ast_sal + accept

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	87	686878				
2	86	595882	1	90996	13.133	0.0004904 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Second Partial F-test

Analysis of Variance Table

Model 1: grad_rate_adj ~ ast_sal + accept

Model 2: grad_rate_adj ~ ast_sal + accept + outstate

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	86	595882				
2	85	489232	1	106649	18.529	4.458e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Third Partial F-test

Analysis of Variance Table

Model 1: grad_rate_adj ~ ast_sal + accept + outstate

Model 2: grad_rate_adj ~ ast_sal + accept + outstate + tuition

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	85	489232				
2	84	439337	1	49896	9.5399	0.002725 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Fourth Partial F-Test

Analysis of Variance Table

Model 1: grad_rate_adj ~ ast_sal + accept + outstate + tuition

Model 2: grad_rate_adj ~ ast_sal + accept + outstate + tuition + top10

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	84	439337				
2	83	436034	1	3302.5	0.6286	0.4301